

SCIENTIFIC REPORT · OBJECTIVE GAIT · BERTONE ET AL. 2026

Greater forelimb symmetry with eCTM.

Blinded, sensor-based smartphone kinematics (Sleip) confirmed veterinary lameness scoring.

Background

OBJECTIVE

Veterinary lameness scoring is inherently subjective and can vary between examiners. In an open-label pragmatic trial we needed an assessor-independent endpoint to corroborate clinical grades. Blinded, smartphone-video kinematic gait analysis (Sleip) quantified gait asymmetry at every visit and tested whether eCTM produced greater symmetry gains than clinician-selected standard of care.

Methods

MEASUREMENT

Trot and lunge videos were captured on smartphones at Weeks 0, 2, 4, and 12 and processed through the Sleip markerless computer-vision pipeline. Two asymmetry indices were derived per limb pair: impact (early stance) and push-off (late stance). Lower asymmetry means a more symmetric gait. Analysts were blinded to treatment.

WHY FORELIMB IS PRIMARY

The forelimb pair is the most reliable Sleip signal in the field — the head-nod reference is well established. Hindlimb data are noisier (no equivalent landmark; recording surfaces vary between practices). Forelimb was therefore the primary kinematic comparison; combined fore+hind data are exploratory.

ANALYSIS

Mixed-effects models compared cumulative change from baseline between eCTM and pooled standard of care at each follow-up. eCTM improvement is expressed as a percentage reduction in asymmetry index vs. standard of care at Week 12.

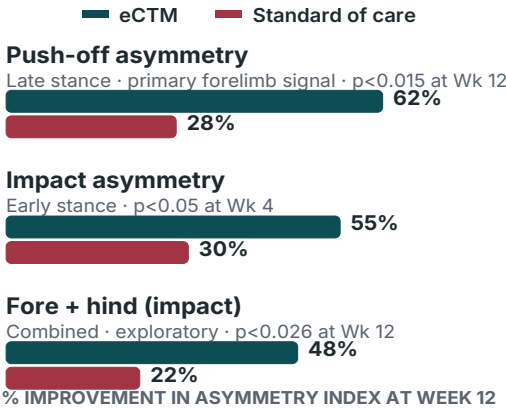
Results

FORELIMB PUSH-OFF ASYMMETRY — IMPROVEMENT VS. STANDARD OF CARE

Wk 2 & Wk 12

Push-off separated at Week 2 ($p<0.003$) and remained ahead at Week 12 ($p<0.015$).

FIG 1 · Cumulative asymmetry improvement at Week 12



eCTM improved sensor-derived asymmetry more than standard of care across all three comparisons (Figure 6 and Supplementary Figure S4).

Interpretation

OBJECTIVE GAIT DATA MATCHES CLINICAL SCORING

A blinded, marker-free kinematic pipeline — independent of the veterinarian who assigned lameness grades — detected the same eCTM advantage: push-off separated at Week 2 ($p<0.003$) and remained ahead at Week 12 ($p<0.015$); impact at Week 4 ($p<0.05$); combined fore+hind impact at Week 12 ($p<0.026$).

Symmetry a sensor could see.

Objective smartphone kinematics corroborated clinical lameness scoring: eCTM horses moved more symmetrically than standard of care across every blinded time point.

Equine Performance Labs

Important information: equicenta® CTM is an investigational veterinary product. Not for human use. Not for use in food-producing animals. The results shown are from a single prospective controlled clinical study in performance horses; individual outcomes may vary. Source: Bertone AL, Davern A, Sutter WW, et al. Front. Vet. Sci. 2026. doi: 10.3389/fvets.2026.1833933 · AVMA VCT #26005934 · IACUC ETCR-2024-0329.